

LANGRES, France

WMO: 072830

Lat: 47.8436N

Lon: 5.3375E

Elev: 467

StdP: 95.84

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.7	-5.9	-11.2	1.5	-5.4	-8.6	1.9	-4.4	11.0	6.6	10.0	5.3	2.8	40	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	30.3	19.5	28.2	18.9	26.4	18.2	20.7	28.0	19.7	26.3	18.9	24.8	3.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.9	23.2	17.4	13.2	22.1	16.7	12.6	21.3	61.5	28.2	58.3	26.4	55.4	25.1	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	7.5	6.5	-9.8	33.4	2.8	2.1	-11.9	34.9	-13.5	36.1	-15.1	37.3	-17.2	38.9		
			-10.3	22.4	2.9	0.8	-12.3	23.0	-14.0	23.4	-15.6	23.9	-17.7	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	1.3	2.4	6.1	10.0	13.1	17.4	19.4	18.5	15.2	10.9	5.8	2.4
	DBStd	7.37	4.01	4.26	4.11	3.98	3.71	3.76	3.66	3.47	3.40	3.95	3.72	3.79
	HDD10.0	1083	269	213	131	48	12	1	0	0	2	38	133	236
	HDD18.3	3111	528	447	378	251	166	61	32	42	106	231	377	494
	CDD10.0	1176	0	1	11	48	109	224	291	262	158	66	7	0
	CDD18.3	160	0	0	0	1	4	34	64	45	12	0	0	0
	CDH23.3	1284	0	0	0	5	32	277	546	348	74	2	0	0
	CDH26.7	371	0	0	0	0	3	75	177	107	8	0	0	0
Wind	WSAvg	3.0	3.5	3.5	3.4	3.0	2.9	2.6	2.6	2.5	2.4	2.9	3.3	3.5
Precipitation	PrecAvg	914	79	65	60	63	84	66	82	74	67	90	94	90
	PrecMax	1143	187	129	172	178	147	119	169	163	141	171	271	239
	PrecMin	645	19	6	1	7	17	29	10	28	15	26	28	7
	PrecStd	129	39	35	42	41	33	26	34	36	37	45	54	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	13.6	18.7	24.1	26.8	31.9	33.7	32.7	27.9	22.8	15.5	11.4
		MCWB	9.8	8.4	11.2	14.9	19.1	20.1	19.8	20.2	18.1	15.7	12.7	9.8
	2%	DB	9.6	11.0	16.6	21.4	23.8	28.8	30.7	29.5	25.6	20.2	13.4	10.2
		MCWB	8.5	7.8	10.1	13.5	16.6	19.4	19.3	19.5	17.7	15.3	11.5	9.2
	5%	DB	8.1	9.3	14.3	19.0	22.0	26.5	28.6	27.1	23.4	18.0	12.1	8.8
		MCWB	7.1	7.0	9.2	12.5	15.7	18.9	18.6	18.7	16.6	14.5	10.7	7.9
	10%	DB	6.7	7.9	12.2	16.8	19.9	24.4	26.4	24.8	21.0	16.0	10.8	7.5
		MCWB	5.9	6.1	8.2	11.2	14.6	17.8	17.8	17.8	15.4	13.6	9.8	6.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	9.5	12.2	15.8	19.9	22.1	21.3	21.6	19.7	17.1	13.5	10.4
		MCDB	10.8	11.9	16.7	22.7	25.9	29.3	29.9	29.6	24.9	20.5	14.6	10.8
	2%	WB	8.8	8.3	10.9	14.2	17.9	20.6	20.3	20.3	18.4	15.9	11.9	9.3
		MCDB	9.5	10.3	14.7	20.2	22.6	26.6	28.2	27.7	23.5	19.0	12.9	10.0
	5%	WB	7.2	7.3	9.8	12.8	16.3	19.3	19.4	19.3	17.3	14.8	10.9	8.0
		MCDB	8.0	9.0	13.3	18.1	20.7	25.2	26.8	25.4	22.0	17.4	12.0	8.8
	10%	WB	6.0	6.3	8.7	11.6	15.0	18.4	18.5	18.4	16.3	13.7	9.7	6.9
		MCDB	6.6	7.7	11.7	16.1	18.9	23.6	25.1	23.6	20.2	15.7	10.7	7.5
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.6	7.7	8.9	8.9	9.7	9.8	9.3	8.9	7.0	4.9	4.4
		MCDBR	5.5	7.9	10.9	11.9	11.7	12.5	12.9	12.8	11.9	9.4	6.4	5.4
		MCWBR	4.7	5.4	6.3	6.5	6.4	5.7	5.1	5.3	5.6	5.3	4.7	4.6
	5% WB	MCDBR	5.1	7.0	9.3	10.9	10.8	11.4	11.7	11.2	10.7	8.4	5.7	5.0
		MCWBR	4.6	5.0	5.8	6.2	6.3	5.7	5.1	5.1	5.3	5.1	4.4	4.4
Clear-Sky Solar Irradiance	taub	0.301	0.322	0.347	0.378	0.385	0.393	0.388	0.384	0.360	0.347	0.317	0.298	
	taud	2.464	2.417	2.361	2.283	2.301	2.310	2.319	2.351	2.396	2.428	2.464	2.454	
	Ebn at Noon	782	839	869	872	877	870	869	857	845	796	752	736	
	Edh at Noon	64	83	103	122	125	125	122	113	99	82	64	58	
All-Sky Solar Radiation	RadAvg	0.98	1.78	3.03	4.37	5.13	5.88	5.60	4.84	3.75	2.27	1.16	0.79	
	RadStd	0.14	0.29	0.41	0.58	0.57	0.55	0.49	0.46	0.32	0.17	0.11	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page